Shrey Painuli

Roll No.: M25EET007
B.Tech - Electrical Engineering
M.Tech - Sensors and Internet of Things
Indian Institute of Technology, Jodhpur

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Github
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology, Jodhpur	-	2025-Present
B.Tech	G.B Pant Institute of Engineering and Technology, Pauri	84.26	2017-2021
Senior Secondary	ICSE Board	82.25	2017
Secondary	ICSE Board	94.40	2015

Experience

- **Chegg India Pvt. Ltd.** *Jan. 2021 - Dec. 2022*
  Subject Matter Expert Work From Home
  – Provided programming solutions to Computer Science students
  – Solved doubts and follow-up queries of Electrical Engineering students

Projects

- **Navigation and Object Identification Aid for the Visually Challenged** *Github*
  B.Tech Major Project, Mentor: Dr. Sanjay Gairola
  – Technologies/Hardware used: TensorFlow, Python and Adafruit libraries, Shell scripting / Raspberry Pi 3B+
  – Objective: Using ultrasonic sensors to gauge the distance of objects, communicating it to the deaf-blind person through a vibration motor. Additionally, employing TensorFlow on the video fed from the camera module to identify objects and communicating the identified objects to the deaf/blind person through tactile/audio feedback.
- **Exigence Band** *Github*
  Two-member Team Project
  – Technologies/Hardware used: I2C and UART protocols / Node MCU, GPS, GSM, Accelerometer and Biosensors
  – Objective: Using biosensors and GPS to keep a regular record of the vitals (BPM, SPO2, Temperature, Location) of the user, which the GSM module sends to the respective contacts and authority upon automatic detection of a fall using accelerometer algorithm or irregular SPO2, or manual press of an emergency button.
- **Kinetic Sand Art Table** *Github*
  Self Project
  – Technologies/Hardware used: GRBL Firmware, SPI protocols / Arduino Uno, Stepper motors
  – Objective: Using Arduino Uno and GRBL firmware to create aesthetic patterns of sand upon the table surface. These patterns are fed manually using G-code, which then drives the stepper motors (XY axis motion) controlling an iron ball with the help of a magnet hidden beneath the surface. This motion of iron ball creates patterns over the pre-leveled sand.

Technical Skills

- **Programming Languages:** : Embedded C, C/C++, Python, Java, BaSh, Verilog, MATLAB*
- **Developer Tools**: Git, Arduino IDE, Keil MDK, Multisim, Xilinx Vivado, Visual Studio, Thonny and Pycharm

** Elementary proficiency*

Achievements

- **International Conference:** Published a research paper in ICASTM-2021 *Dec. 2021*
- **GATE:** Secured an AIR-6021 in Electrical Engineering *Mar. 2025*
- **Merit Scholarship:** Secured college scholarship given to the Top 5 students of EE Dept.
- **College Fest:** Begged top positions in various competitions securing third place in Mr. Goonj (College Fest Title)

Extracurriculars

- **Literary and Editorial Head:** Held this position in the Society of Electrical Engineering (G.B.P.I.E.T) *2018-19*
- **Sports Fest:** Won the inter-branch basketball competition *2019*
- **Technical Quiz:** Third place in the quiz organised by Society of Electrical Engineering (G.B.P.I.E.T) *2017*
- **International Robotics League:** Built powerful robots to participate in the Robo Soccer event *2019*
- **Internship:** Six weeks training in Ensino Research And Development Pvt. Ltd, on real time IoT projects *2020*
- **Development Program:** Participated in an IoT FDP organised by AIT, Bengaluru *2020*

Publications

- **Painuli, S.,** Gairola, S., Arya, K., Tiwari, D., Balram. (2021). Navigation and Object Identification Aid for the Visually Challenged. SAMRIDDHI : A Journal of Physical Sciences, Engineering and Technology, Volume 13, Special Issue (2), 168-173 *Research Publication*